

Unit 9, Lesson 2: Writing a Quadratic Function From a Graph



Benchmarks

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.

MA.912.AR.3.4 Write a quadratic function to represent the relationship between two quantities from a graph, a written description, or a table of values within a mathematical or real-world context.



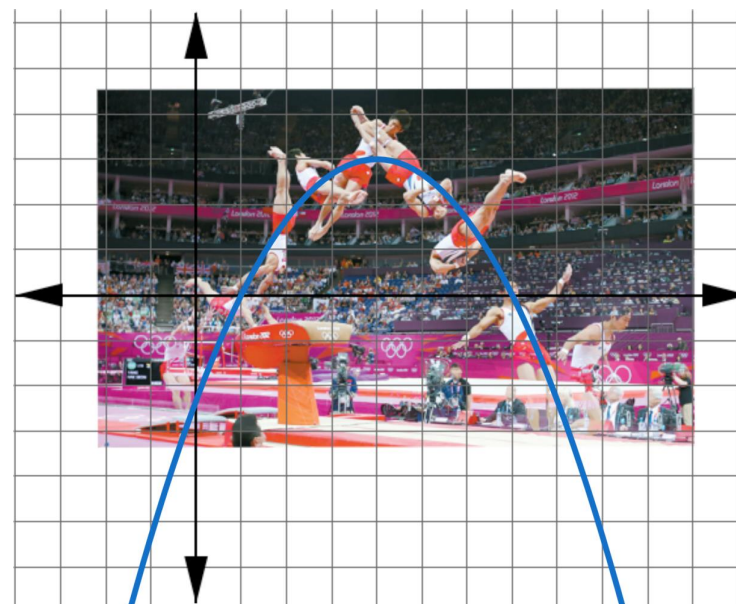
Learning Targets

- ☐ I can write a quadratic function to represent the relationship between two quantities from a graph.
- ☐ I can rewrite a quadratic equation from vertex form to standard form.
- ☐ I can solve for a variable of interest.



9.2.1 Warm-Up: Vaulting Moves

Vaulting is an exercise in which gymnasts leap over an apparatus and perform a skill in mid-air. The image shows a frame-by-frame pictorial of a gymnast's vault routine. Based on the compilation of moves performed by the gymnast, a sketch of a parabola was drawn.



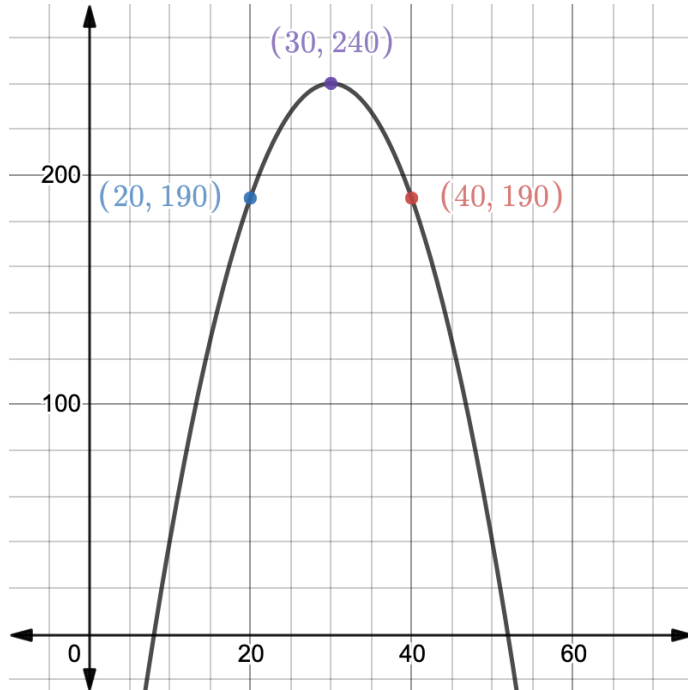
1. Explain whether the a value of the parabola is positive or negative.
2. If each square is 1 by 1, what are the vertex and x -intercepts of the parabola?

vertex	
x -intercepts	



9.2.2 Guided Instruction: Writing a Quadratic Function in Vertex Form From a Graph

Hudson is starting a lawn-mowing business in his neighborhood. He creates a graph to help him determine what to charge customers per mow to maximize his profits. He uses x to represent the cost of each lawn he mows and y to represent the profit in dollars (\$).



1. The point
 - ☐ (20, 190)
 - ☐ (30, 240)
 - ☐ (40, 190) represents the maximum profit Hudson can make. This point on a graph is called the _____ of the parabola.

2. Discuss with your partner what you notice about points (20, 190) and (40, 190).
3. Points (20, 190) and (40, 190) are equidistant from the vertex (30, 240). Identify two other points on the graph that are equidistant from the vertex (30, 240).
4. To find the equation of the quadratic function represented by the graph, it would be easiest to use
- ☐ factored form
 - ☐ standard form
 - ☐ vertex form
- because the graph provides the _____ and at least one other _____.

The steps in the table show how to use the point (20, 190) and the vertex (30, 240) to find the quadratic function in vertex form $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex.

Substitute values for the vertex (30, 240).	$f(x) = a(x - 30)^2 + 240$
Substitute values for the point (20, 190).	$190 = a(20 - 30)^2 + 240$
Solve for the variable a .	$190 = a(-10)^2 + 240$ $190 - 240 = a(100)$ $-50 = a(100)$ $a = -\frac{1}{2}$
Substitute in the vertex and a value.	$f(x) = -\frac{1}{2}(x - 30)^2 + 240$

The quadratic equation Hudson graphed is $y = -\frac{1}{2}(x - 30)^2 + 240$.



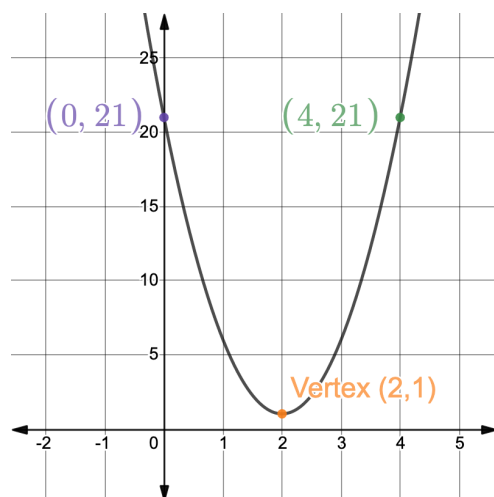
9.2.3 Collaboration: Writing a Quadratic Function Using a Vertex and Point From a Graph

- With your partner, use the other point identified on Hudson's graph, (40, 190), along with the same vertex, (30, 240), to generate another quadratic equation in vertex form.

Substitute values for the vertex (30, 240).	$f(x) = a(x - 30)^2 + 240$
Substitute values for the point (40, 190).	$190 = a(40 - 30)^2 + 240$
Solve for the variable a .	
Substitute in the vertex and a value.	

- The quadratic equation with vertex (30, 240) and point (40, 190) is _____.
- Discuss with your partner what you notice about quadratic functions with the same vertex and two points that are equidistant from the vertex. Summarize your discussion.

2. A quadratic function is graphed with the vertex and two points equidistant from the vertex labeled.



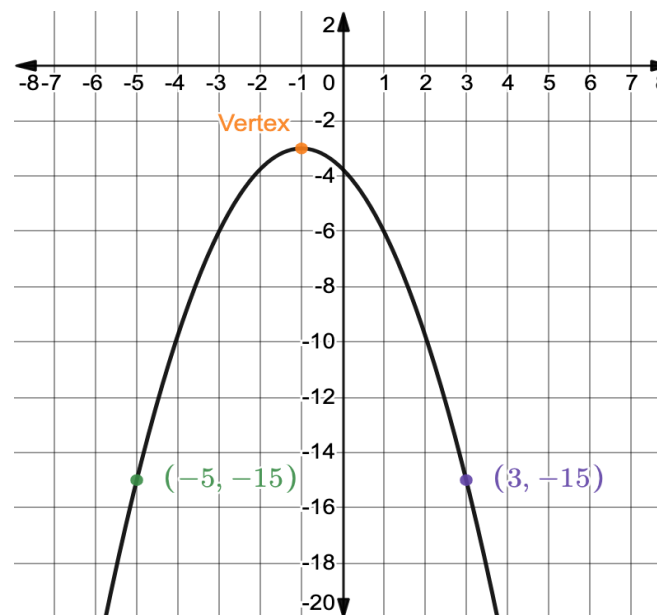
- a. Write the equation in vertex form for the quadratic function graphed.

Substitute values for the vertex (2, 1).	
Substitute values for the point (0, 21) or (4, 21).	
Solve for the variable a .	
Substitute in the vertex and a value.	

- b. The quadratic equation is _____.

- c. Rewrite the equation into standard form
 $y = ax^2 + bx + c$.

3. Write the quadratic equation for the graph.

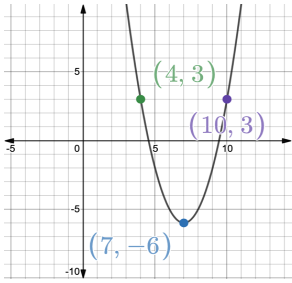


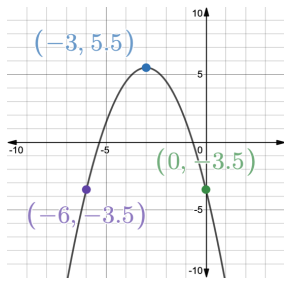
Vertex Form	
Standard Form	

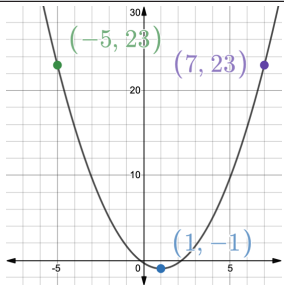


9.2.4 Your Turn!

1. Complete the table for each quadratic function using the graph of the function.

Graph A	Vertex and Point	Vertex Form
	Vertex:	
	Point:	Standard Form

Graph B	Vertex and Point	Vertex Form
	Vertex:	
	Point:	Standard Form

Graph C	Vertex and Point	Vertex Form
	Vertex:	
	Point:	Standard Form



9.2.5 Wrap-Up: Reflection

1. How “in the zone” do you feel right now about your understanding of writing equations of quadratic functions?

